



Cooperative Scheduling of Aerial and Ground Service Vehicles for Urban Public Service Tasks: A Multi-Objective Optimization Approach



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Abstract: Efficient coordination of heterogeneous mobile resources is essential for delivering large-scale urban services, such as sanitation, infrastructure inspection, or last-mile delivery. This study addresses the problem of scheduling aerial and ground service vehicles—unmanned aerial vehicles (UAVs) and mobile ground crews—to cover spatially distributed demand points under operational constraints. We formulate the task as a multi-objective optimization problem that simultaneously maximizes service coverage, minimizes total completion time, and optimizes resource utilization while respecting safety, capacity, and time-window restrictions. A hierarchical solution framework is proposed: global task allocation first assigns demand zones to vehicle types according to their capabilities, and local path planning then generates efficient routes for each agent. A dynamic re-optimization mechanism adjusts schedules in real time when disturbances occur, such as resource depletion or environmental changes. The method is evaluated on scenarios of increasing scale (51, 113, and 212 demand points) that emulate urban public spaces. Results from ten repeated experiments show that the cooperative strategy achieves coverage rates (CRs) above 97% across all scales, reduces total operation time (TOT) by up to 33% compared with single-mode operations, and improves resource efficiency by 21.10% and 47.40%. Statistical analysis confirms the robustness of the improvements. The framework offers a scalable, resource-aware solution for coordinating heterogeneous service fleets, with direct applicability to intelligent transportation systems, particularly in demand-responsive urban services and multimodal fleet management.

Keywords: Urban service systems; Service vehicle scheduling; Aerial-ground coordination; Demand-responsive mobility; Multi-objective optimization; Coverage path planning; Task allocation

1 Introduction

The recurrent outbreaks of public health emergencies have made large-scale disinfection of public spaces a critical component of urban governance and public safety. In high-traffic and structurally complex environments—such as transportation hubs, shopping malls, plazas, and multi-story public buildings—timely and comprehensive disinfection plays a vital role in interrupting airborne and surface-based transmission of pathogens, mitigating cross-infection risks, and maintaining public confidence in governmental health management systems.

From an operational research perspective, coordinating multiple mobile units to cover spatially distributed service points under resource and time constraints is a classic problem in intelligent transportation systems (ITS) [1]. Applications range from parcel delivery and transit fleet management to emergency response and urban infrastructure inspection. This study addresses a particular instance of such problems—large-scale sanitation operations in public spaces—where heterogeneous service vehicles, including unmanned aerial vehicles (UAVs) and ground crews, must be dispatched efficiently to meet coverage requirements. The task allocation, path planning, and dynamic scheduling

challenges involved are fundamentally aligned with core challenges in ITS, including demand-responsive mobility, multimodal freight transport, and urban service fleet management.

Conventional manual disinfection remains widely used for small-scale or preventive operations; however, it is constrained by low efficiency, incomplete coverage, high labor intensity, and increased exposure risks. In contrast, UAV-based spraying or fogging systems have demonstrated rapid deployment and wide-area coverage in open environments [2], confirming their effectiveness for large-scale sanitation operations [3, 4]. Nevertheless, UAV-only operations face inherent limitations, including restricted payload capacity, limited endurance, and reduced robustness when operating in obstacle-dense or dynamically changing environments, where safe motion planning and obstacle avoidance become critical challenges [5–8]. Ground operators or mobile robotic platforms, on the other hand, offer superior maneuverability and accuracy in confined or cluttered spaces and can provide high-precision supplementary disinfection [9, 10]. Such ground-based operations typically rely on reliable perception and localization capabilities to ensure robustness in dynamic public environments with frequent human activity [11]. Sole reliance on ground-based disinfection, however, substantially increases total operation time (TOT), human-resource consumption, and exposure risks in complex public environments [12].

These trade-offs highlight the necessity of integrating UAV-based rapid coverage with ground-based precision operations to achieve balanced performance in efficiency, coverage completeness, resource utilization, and operational safety. From a systems perspective, such integration inherently requires coordinated task allocation, dynamic motion planning, and cooperative control among heterogeneous agents, rather than isolated single-platform optimization [13].

In recent years, extensive research has been devoted to coverage path planning (CPP) [14], energy-efficient and cooperative multi-UAV operations [15], and time-minimizing full-area coverage strategies [16]. Related studies have also explored indoor and three-dimensional disinfection robotics [17, 18], meta-heuristic optimization methods for complex planning problems [19], and large-scale robotic disinfection applications [20]. Meanwhile, significant progress has been achieved in real-time motion planning and decision-making for mobile robotic systems operating in unknown or cluttered environments. For instance, Yang et al. [21] proposed the RAMPAGE framework to enable whole-body, agile motion planning for mobile manipulators under real-time constraints, offering valuable insights into dynamic planning strategies applicable to ground-based sanitation operations in complex public spaces.

Representative application-oriented studies further demonstrate the effectiveness of robotic disinfection and coverage optimization. Bezas et al. [22] proposed a point-of-interest detection and coverage algorithm for UAV swarms and analyzed the influence of path patterns on operational efficiency. Xu et al. [23] designed a three-dimensional disinfection robot and developed a CPP algorithm tailored to complex environments. Peñacoba et al. [24] compared bio-inspired meta-heuristic algorithms for ultraviolet-C disinfection route optimization. Jorge et al. [3] conducted both field experiments and simulations on UAV spraying operations in outdoor public spaces. Wang et al. [25] proposed a low-complexity, map-based path optimization method that significantly reduced path length while improving surface coverage.

Despite these advances, research on collaborative disinfection planning involving UAVs and ground personnel remains limited. Existing studies predominantly focus on single-platform optimization and lack: (1) a unified multi-objective optimization framework that simultaneously accounts for coverage effectiveness, TOT, route utilization efficiency (RUE), personnel workload, and safety constraints; (2) systematic task-allocation and dynamic adjustment mechanisms for coordinating UAVs and ground agents under varying environmental and operational conditions; and (3) integrated scheduling and safety-management strategies specifically designed for complex public environments such as transportation hubs, shopping centers, and large open plazas.

Compared with traditional heuristic-based scheduling methods and metaheuristic optimization approaches widely used in emergency response systems, the proposed framework adopts a hierarchical multi-objective structure that explicitly integrates coverage, time efficiency, resource utilization, and safety constraints within a unified model. Unlike single-objective or loosely coupled optimization strategies, the proposed method simultaneously addresses task allocation and path planning under dynamic conditions, enabling more comprehensive coordination between UAVs and ground personnel. This integrated design improves scalability and adaptability in large-scale public-space scenarios.

Although existing studies have explored UAV-based service delivery or ground-based scheduling strategies, most approaches focus on single-resource optimization and lack an integrated coordination mechanism. Moreover, few studies address hierarchical multi-objective optimization and dynamic re-optimization under resource variation, which limits their adaptability in complex public-space scenarios.

Beyond its public health motivation, the problem investigated here contributes to the broader ITS literature on heterogeneous fleet coordination. The joint optimization of coverage, time, and resource utilization under dynamic conditions mirrors challenges in real-time ride-sharing, last-mile delivery with drones, and multi-priority service scheduling. Therefore, the proposed framework not only provides a solution for sanitation tasks but also offers methodological insights applicable to other ITS domains.

To address these gaps, this manuscript proposes a multi-objective optimization method for collaborative UAV-

ground-based service operations planning and scheduling in the public health management of public places. The proposed approach jointly optimizes collaboration efficiency (CE), TOT, RUE, personnel workload, and safety constraints. A task-allocation strategy is developed in which UAVs perform rapid large-area coverage while ground personnel execute precise supplementary disinfection in confined or obstacle-dense regions. Moreover, dynamic path-planning and collaborative scheduling algorithms are incorporated to enhance adaptability to complex and evolving environments. Field experiments conducted in representative public spaces, with comparisons against UAV-only, ground-only, and conventional manual disinfection strategies, demonstrate significant improvements in coverage completeness, execution efficiency, and overall resource utilization.

Based on the analysis of existing research on UAV-based coverage optimization, ground-based task scheduling, and multi-agent coordination, the main technical contributions of this paper are summarized as follows:

(1) **Unified Multi-Objective Optimization Framework:** A comprehensive optimization model is constructed to simultaneously consider CE, TOT, RUE, workload balance, and safety constraints within a single formulation, enabling coordinated performance improvement across multiple dimensions.

(2) **Hierarchical Collaborative Scheduling Mechanism:** A layered architecture is designed to decouple global task allocation from local path planning, allowing scalable coordination between multiple UAVs and ground personnel in large-scale public environments.

(3) **Dynamic Incremental Re-Optimization Strategy:** An adaptive scheduling mechanism is introduced to support real-time task adjustment under environmental changes, resource fluctuations, and agent status updates, enhancing system robustness and operational continuity.

(4) **Resource-Aware and Scalable Design:** UAV battery levels, disinfectant supply, and personnel workload constraints are explicitly integrated into the optimization model, ensuring practical feasibility and scalability in complex scenarios.

Compared with existing UAV-only or ground-only approaches, the proposed framework provides a more integrated, adaptive, and resource-efficient solution for collaborative disinfection planning in public spaces.

2 Problem Formulation and Optimization Methodology

2.1 Task and Area Modeling

Public sanitation operations typically involve diverse environments such as transportation hubs, urban squares, commercial complexes, and multi-story indoor spaces. To enable efficient task management, these environments must first be partitioned into manageable sub-areas. The partitioning process considers not only physical boundaries but also pedestrian flow, functional attributes, and potential risk levels, thereby allowing each sub-area to be assigned an appropriate disinfection priority. High-priority zones—those with dense foot traffic or elevated infection risk—are scheduled first, whereas low-risk zones can be treated later to optimize the allocation of resources.

Sanitation operations are further constrained by strict time windows to avoid conflicts with public activities. For example, outdoor squares or transportation hubs are generally disinfected during off-peak periods or temporary closures. Such time constraints ensure operational feasibility, minimize disturbance to the public, and improve overall efficiency.

Within this framework, the disinfection operation can be abstracted as a spatio-temporal coverage problem. Each sub-area is characterized by spatial data (e.g., area size, corridor width, and obstacle distribution), disinfection priority, and an allowable operating time window. These parameters provide quantitative inputs for UAV and ground-crew path planning. Coverage metrics can be defined such that UAVs perform rapid, large-scale sweeps, while ground personnel carry out fine-grained treatment in complex or high-priority zones.

This abstraction transforms a complex public-space disinfection campaign into a well-defined scheduling problem, supplying clear task-allocation rules and operational constraints. As a result, it establishes the foundation for subsequent cooperative scheduling and multi-objective optimization, enabling simultaneous improvements in CE, operational efficiency, and disinfectant utilization.

To ensure methodological transparency and practical reproducibility, the proposed collaborative optimization framework is implemented through a structured hierarchical procedure. The solution integrates task allocation, route optimization, and dynamic adjustment into a coordinated iterative process. The detailed procedure is described as follows.

Step 1: Initialization

All relevant parameters are initialized, including the task regions to be disinfected, the available UAVs, and the ground personnel resources. Operational attributes such as disinfectant capacity, battery endurance, travel speed, safety requirements, and time constraints are specified. Based on these inputs, a feasible initial assignment is generated to satisfy basic coverage and operational feasibility conditions, providing a starting solution for further refinement.

Step 2: Multi-Objective Model Establishment

The collaborative scheduling problem is formulated by jointly considering multiple operational objectives, including improvement of coverage efficiency, reduction of total execution time, balanced workload distribution, and efficient utilization of disinfectant resources. These objectives are integrated into a unified optimization framework subject to operational, safety, and capacity constraints. The model aims to coordinate heterogeneous resources while maintaining feasibility under realistic execution conditions.

Step 3: Hierarchical Task Allocation

To improve scalability and reduce computational burden, a hierarchical allocation mechanism is adopted. Tasks are first classified according to spatial distribution and environmental characteristics. Based on operational suitability, tasks that benefit from aerial coverage are preferentially assigned to UAVs, whereas tasks requiring precise maneuvering or obstacle avoidance are allocated to ground personnel. This decomposition strategy reduces the complexity of direct joint optimization and enhances computational efficiency.

Step 4: Route Optimization

After task assignment, route optimization is conducted for each UAV and ground operator. The objective is to minimize travel time and unnecessary detours while respecting operational constraints. An iterative refinement mechanism is used to improve routing sequences, ensuring that collaborative execution remains efficient and balanced across all participating resources.

Step 5: Dynamic Adjustment Mechanism

To enhance adaptability in dynamic environments, a real-time adjustment mechanism is incorporated. When unexpected events occur—such as UAV battery reduction, personnel availability changes, or environmental disturbances—the system updates relevant operational parameters and performs localized re-optimization. This strategy allows the framework to respond efficiently without recomputing the entire schedule, thereby maintaining overall system stability.

Step 6: Termination and Output Generation

The iterative optimization process continues until no significant performance improvement is observed or a predefined iteration limit is reached. The final output consists of coordinated task allocation plans for UAVs and ground personnel, along with optimized routing schedules that collectively ensure efficient and balanced sanitation operations.

2.2 Resource Modeling

2.2.1 Interdependency of resource variables

In the collaborative disinfection framework, resource variables associated with UAVs, ground personnel, and disinfectant supply are not independent but exhibit strong interdependencies within the optimization process.

For UAVs, battery level, payload capacity, flight distance, and disinfectant consumption are mutually constrained. Increased task distance directly affects energy consumption, which in turn limits available payload for disinfectant spraying. Therefore, task allocation decisions must simultaneously consider energy usage and spraying requirements to ensure mission feasibility.

For ground personnel, workload, travel time, and disinfectant usage are interrelated. Longer travel routes increase total working time, which must remain within safety limits. Meanwhile, disinfectant consumption affects the remaining task capacity and may trigger reallocation if supply thresholds are reached.

Additionally, disinfectant supply acts as a shared resource constraint linking UAV and ground operations. The total consumption of both platforms must satisfy global availability limits, ensuring coordinated resource utilization across the entire system.

These interdependencies are explicitly embedded into the multi-objective optimization model through coupled constraint equations, enabling joint consideration of operational efficiency, coverage performance, and resource feasibility.

2.2.2 Resource constraint formulation

In collaborative sanitation operations for public places, the primary operational resources include UAVs, ground-based service operations personnel, and disinfectant supplies. Proper modeling of these resources provides a quantitative basis for task allocation and path planning, and establishes the constraints required for subsequent multi-objective optimization.

UAV resources include the number of units, battery endurance, flight speed, maximum payload, and disinfectant capacity. During sanitation operations, UAVs must balance coverage efficiency and flight safety. Battery endurance and payload capacity determine the maximum area a UAV can cover in a single mission, while flight speed and altitude constraints affect both operation time and disinfection effectiveness. Therefore, task allocation and path planning must comprehensively consider these physical performance constraints to ensure task completion and optimal resource utilization.

Key UAV constraints include the number of units, maximum flight time, maximum payload, and disinfectant capacity. Let the set of UAVs be $U = \{u_1, u_2, \dots, u_m\}$.

Each UAV u_i has a maximum flight time $T_i^{\max}$ and a maximum payload $W_i^{\max}$, with a disinfectant capacity C_i . During a mission, the flight time t_i and disinfectant consumption d_i of UAV must satisfy:

$$t_i \leq T_i^{\max}, w_i \leq W_i^{\max}, d_i \leq C_i, \forall i \in U \quad (1)$$

where, w_i denotes the current payload of UAV u_i , and d_i represents its current disinfectant consumption. UAVs are primarily responsible for large-area coverage tasks, and their path planning must ensure full coverage while simultaneously satisfying flight safety and resource constraints.

Ground-based service operations personnel resources include the number of staff, workload, movement speed, and safety constraints. Ground personnel are mainly responsible for precise disinfection in complex environments such as stairways, narrow corridors, or areas with dense obstacles. During task allocation, work areas must be assigned reasonably to avoid excessive workload for any single operator and to ensure that high-priority areas receive timely disinfection. In addition, personnel safety constraints must be considered to prevent exposure to high-risk substances or hazardous zones.

Let the set of ground personnel be $G = \{g_1, g_2, \dots, g_n\}$. Each operator g_k has a workload l_k that must satisfy the safety limit $L_k^{\max}$ and complete the assigned tasks within the allowable working time $T_k^{\max}$. The path constraint can be expressed as:

$$\sum_{(j,m) \in P_k} t_{jm} \leq T_k^{\max}, l_k \leq L_k^{\max}, \forall k \in G \quad (2)$$

where, P_k represents the set of task points along the operator's path, and t_{jm} is the travel time between points j and m . By reasonably assigning work areas and routes, workload balance can be achieved while ensuring that high-priority areas are treated first.

As a core consumable, disinfectant capacity, replenishment methods, and usage efficiency directly affect the continuity and economy of operations. UAVs and ground personnel must consider remaining disinfectant levels and replenishment strategies during path planning to ensure that tasks can be completed without running out of supplies. The disinfectant's spraying efficiency is also related to coverage, so it should be treated as both a constraint and an optimization objective.

Let the total available disinfectant be D^{total} . The total consumption must satisfy:

$$\sum_{i \in U} d_i + \sum_{k \in G} d_k \leq D^{\text{total}} \quad (3)$$

where, d_i and d_k represent the disinfectant consumption of UAV u_i and ground operator g_k , respectively.

By modeling UAVs, ground personnel, and disinfectant resources comprehensively, the quantity, constraints, and roles of various resources are clearly defined, providing fundamental data for collaborative scheduling. On this basis, subsequent path planning and task allocation algorithms can achieve multi-objective optimization, balancing operational efficiency, coverage, and resource utilization to form a complete collaborative disinfection operation model.

2.3 Collaborative Service Operations Model

(1) Collaborative Task Allocation

In large-scale urban service operations, UAVs and ground personnel must cooperate complementarily to balance the need for rapid wide-area coverage and precise execution in complex zones. The collaborative service model specifies the division of labor among operational agents, path-planning constraints, and conflict-avoidance mechanisms to optimize overall task efficiency and resource utilization.

UAVs are primarily responsible for rapid coverage of large open regions, whereas ground personnel handle high-risk, structurally complex, or UAV-inaccessible areas. Let the total task-region set be $R = \{r_1, r_2, \dots, r_N\}$, the UAV-assigned region set be $R_U \subset R$, and the ground-crew region set be $R_G \subset R$. The task allocation satisfies

$$R_U \cup R_G = R, R_U \cap R_G = \emptyset \quad (4)$$

which ensures that UAVs and ground personnel do not redundantly disinfect the same area, avoiding resource waste.

(2) Time-Window Constraint

Operations in public venues must adhere to prescribed time windows to prevent interference with public activities. Let the allowable operation time of region r_j be $TW(r_j) = [t_j^{\text{start}}, t_j^{\text{end}}]$. Denote the working times of UAV u_i and ground worker g_k in region r_j as t_{ij} and t_{kj} , respectively. The time-window constraint is

$$t_j^{\text{start}} \leq t_{ij} \leq t_j^{\text{end}}, t_j^{\text{start}} \leq t_{kj} \leq t_j^{\text{end}} \quad (5)$$

(3) UAV Path-Planning Constraints

UAV path planning must maximize coverage while satisfying obstacle-avoidance and flight-safety requirements. Define the coverage ratio of UAV u_i over region $r_j \in R_U$ as $c_{ij} \in [0, 1]$. The overall UAV coverage can be expressed as

$$C_U = \frac{\sum_{i \in U} \sum_{r_j \in R_U} c_{ij}}{|R_U|} \quad (6)$$

The flight path of UAV u_i , denoted p_i , must also avoid obstacles and restricted zones, which can be enforced by incorporating a constraint function $f_{obs}(p_i) = 0$ into the planning algorithm.

(4) Ground Personnel Route Planning and Dynamic Load Balancing

Ground personnel must complete tasks while ensuring safety and balanced workloads. Let the route of ground worker g_k be P_k and the set of regions covered along this route be $R_k \subseteq R_G$. The workload-balancing and priority constraints are

$$l_k = \sum_{r_j \in R_k} L(r_j) \leq L_k^{\max}, \forall k \in G \quad (7)$$

where, $L(r_j)$ represents the workload of region r_j , $P(r_j)$ is the priority level of region r_j , and τ denotes the high-priority threshold, $P(r_j) > \tau$.

(5) Redundant Disinfection and Conflict Avoidance

To ensure efficient resource utilization, UAVs and ground personnel must avoid duplicating disinfection over the same region. Let a region r_j already be covered by UAVs; the ground personnel's priority for that region is reduced or skipped:

$$f_{c_{ij}} \geq \theta, r_j \notin R_k \quad (8)$$

where, θ is the coverage threshold. Regions exceeding this threshold do not require additional disinfection by ground personnel.

(6) Spatial Safety Constraint

UAV flight paths must avoid obstacles and restricted zones while maintaining a safe distance d_{safe} :

$$\text{dist}(p_i(t), O) \geq d_{\text{safe}}, \forall i \in U \quad (9)$$

where, $p_i(t)$ denotes the position of UAV u_i at time t , and O is the set of obstacles. This constraint ensures safe UAV operations and prevents collisions.

(7) Collaborative Coverage Constraint

The collaborative coverage efficiency of UAVs and ground personnel can be represented using coverage matrices $C = [c_{ij}]$ and $G = [g_{kj}]$, with the requirement that overall coverage satisfies:

$$\sum_{r_j \in R} \max \left(\sum_{i \in U} c_{ij}, \sum_{k \in G} g_{kj} \right) \geq C_{\min} \cdot |R| \quad (10)$$

where, $C_{\min}$ is the minimum coverage requirement for the task region, ensuring that critical areas are not missed.

Through the modeling of these collaborative constraints, UAV and ground personnel task allocation, path planning, workload balancing, coverage conflict avoidance, and safety requirements are all quantified, forming a complete collaborative disinfection constraint system. This provides explicit constraints for subsequent multiobjective optimization methods, effectively guiding the design and implementation of collaborative scheduling strategies.

2.4 Objective Functions

2.4.1 Joint optimization and objective coupling mechanism

The multiple objectives defined in this study are inherently interrelated rather than independent. CE, TOT, RUE, workload balance, and safety constraints are coupled through shared decision variables, including task allocation, path planning, and resource usage.

For example, increasing coverage may lead to longer operation time, while reducing operation time may affect RUE. Therefore, these objectives exhibit trade-offs that must be balanced within a unified optimization framework.

To address this coupling, the proposed model formulates a joint multi-objective optimization problem in which all objectives are optimized simultaneously under shared constraints. A weighted aggregation strategy is adopted to transform the multi-objective problem into a unified optimization formulation, enabling coordinated improvement across different performance dimensions.

Through this joint optimization mechanism, the system avoids isolated objective tuning and ensures that improvements in one metric do not violate feasibility or degrade other critical performance indicators.

2.4.2 Multi-objective optimization strategy

In collaborative sanitation operations within public spaces, the objective functions must comprehensively consider operational coverage, total execution time, disinfectant usage efficiency, and operational safety. Clear mathematical formulations provide quantitative metrics for subsequent multi-objective optimization.

(1) Maximization of Service Coverage

Service coverage is a critical indicator of task completion. Let the total task region set be $R = \{r_1, r_2, \dots, r_N\}$, UAV u_i coverage ratio for region r_j be c_{ij} , and ground personnel g_k coverage ratio be g_{kj} . Then, the overall disinfection coverage can be expressed as:

$$C_{\text{total}} = \frac{1}{N} \sum_{r_j \in R} \max \left(\sum_{i \in U} c_{ij} \sum_{k \in G} g_{kj} \right) \quad (11)$$

The objective is to maximize C_{total} , ensuring effective coverage of all areas.

(2) Minimization of TOT

The TOT includes the execution time of both UAVs and ground personnel, defined as:

$$T_{\text{total}} = \max \left(\max_{i \in U} t_i, \max_{k \in G} t_k \right) \quad (12)$$

where, t_i and t_k denote the time required by UAV u_i and ground personnel g_k to complete their assigned tasks, respectively. By optimizing path planning and task allocation, TOT can be minimized, improving disinfection efficiency.

(3) Maximization of Disinfectant Usage Efficiency

Disinfectant usage efficiency reflects the economic and environmental aspects of resource utilization. It is defined as:

$$E_{\text{use}} = \frac{\sum_{r_j \in R} r_j}{\sum_{i \in U} d_i + \sum_{k \in G} d_k} \quad (13)$$

where, r_j is the effective disinfected area, and d_i and d_k are the disinfectant consumption of UAV u_i and ground personnel g_k , respectively. Maximizing this metric reduces resource waste and enhances the cost-effectiveness of sanitation operations.

(4) Operational Safety Constraints

During optimization, the safety of personnel must be ensured, avoiding overload or risks in high-priority areas. Safety constraints can be expressed as:

$$l_k \leq L_k^{\max}, t_k \leq T_k^{\max}, \forall k \in G \quad (14)$$

Additionally, UAV flight paths and operational areas must satisfy obstacle avoidance and minimum safe distance constraints, ensuring overall operational safety.

(5) Integrated Multi-Objective Function

In summary, the multi-objective optimization problem can be formulated as:

$$\max (C_{\text{total}}, E_{\text{use}}), \min (T_{\text{total}}) \quad (15)$$

Subject to all resource constraints, task assignment constraints, and safety constraints. In practical optimization, methods such as weighted sum, Pareto optimality, or multi-objective evolutionary algorithms can be applied to achieve comprehensive optimization of UAV and ground personnel collaborative sanitation operations in terms of efficiency, coverage, and resource utilization.

To address the inherent trade-off among multiple objectives, a weighted joint optimization strategy is adopted. Coverage maximization, operation time minimization, and disinfectant efficiency improvement are optimized simultaneously under a unified framework. High-priority regions are assigned larger weights to ensure precision, while global time-related objectives are used to improve operational efficiency. This design enables adaptive balancing between accuracy and efficiency in complex environments.

2.5 Optimization Methodology

2.5.1 Method overview

This study addresses the joint task scheduling problem for UAVs and ground-based service operations personnel by proposing a multi-objective optimization framework. The overall approach abstracts the collaborative aerial-ground operation into a multi-objective optimization problem, with key performance indicators including total task

duration, energy consumption, CE, and execution cost. Optimal scheduling solutions that balance operational efficiency and resource utilization are obtained through iterative algorithmic processes.

In the modeling phase, the operational regions, resource characteristics, and task requirements of both UAVs and ground personnel are uniformly described to ensure computational tractability and model scalability. Subsequently, a multi-objective optimization approach is introduced as the solution engine, allowing flexible selection among typical strategies such as genetic algorithms, particle swarm optimization, or reinforcement learning, depending on the real-time and precision requirements of different scenarios.

During algorithm execution, a hierarchical collaborative task allocation strategy is employed, considering task priorities and dynamic environmental changes. At the global level, the focus is on overall path planning and resource allocation, while at the local level, real-time sensor data and feedback are used to dynamically adjust assignments, enabling efficient coordination between UAVs and ground personnel. This method not only balances multi-dimensional objectives in complex environments but also provides a unified technical framework for subsequent algorithmic refinement and practical deployment.

The proposed framework follows an online monitoring and incremental re-optimization strategy, enabling adaptive task adjustment under dynamic environmental conditions.

To enhance the clarity and transparency of the proposed optimization framework, an overall system architecture diagram is provided to illustrate the structural organization and interaction mechanism of the multi-objective collaborative optimization model. As shown in the Figure 1, the framework integrates task-region inputs, heterogeneous resource constraints, and environmental information into a unified optimization process. The model is composed of three core layers: (1) resource and safety constraint modeling; (2) multi-objective collaborative optimization; (3) hierarchical optimization engine for dynamic scheduling and incremental adjustment. The optimized outputs include UAV task allocation, ground personnel route planning, and collaborative scheduling strategies. This layered architecture explicitly demonstrates how the optimization objectives, constraints, and decision variables are systematically integrated, thereby forming a closed-loop collaborative decision-making mechanism suitable for complex public-space disinfection scenarios.

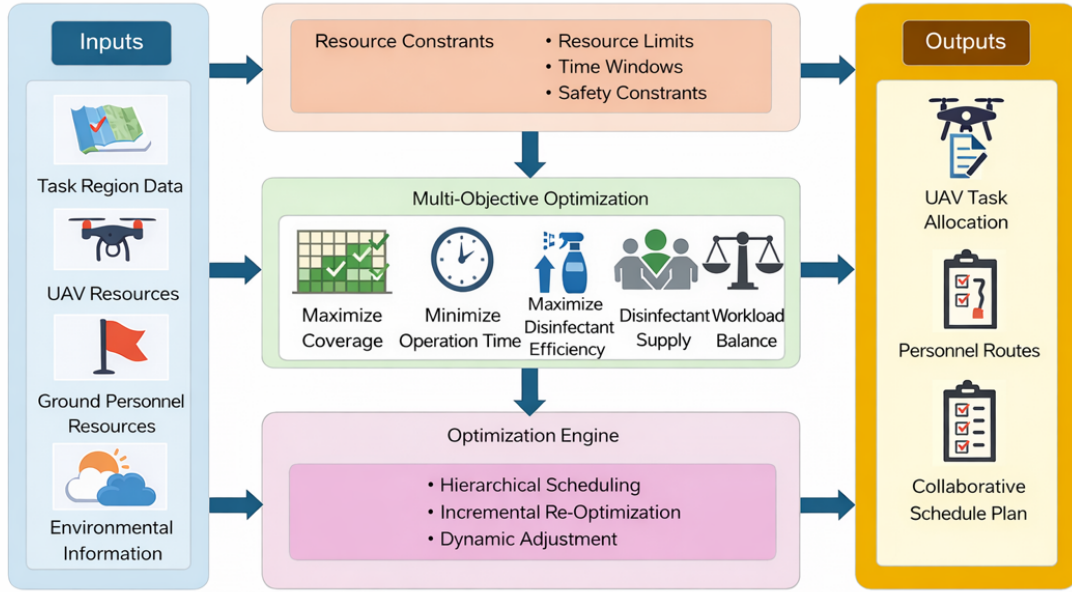


Figure 1. Overall architecture of the proposed multi-objective collaborative optimization framework

2.5.2 Unmanned aerial vehicle path planning

In collaborative sanitation operations, the primary objective of UAV path planning is to achieve full coverage of the designated area with minimal flight distance and time while ensuring operational safety and stability. To this end, the task area can be discretized into a set of grids $R = \{r_1, r_2, \dots, r_N\}$, and the planning problem is formulated as a coverage path optimization model. Let the UAV path be represented by a sequence of nodes $N = \{n_1, n_2, \dots, n_K\}$, where each node n_k corresponds to a grid or waypoint. The objective function can be expressed as:

$$\min_N D(N) = \sum_{k=1}^{K-1} w_{k,k+1} d(n_k, n_{k+1}) \quad (16)$$

where, $d(n_k, n_{k+1})$ denotes the weighted distance considering wind speed between two waypoints, and $w_{k,k+1}$ is the corresponding weight for energy or flight cost. To ensure complete coverage of the area, the following coverage constraint must be satisfied:

$$\bigcup_{n_k \in N} c(n_k) \supseteq R \quad (17)$$

where, $c(n_k)$ represents the spraying coverage range of waypoint n_k .

During flight, UAVs must perform real-time obstacle avoidance and adapt to environmental changes. Let the UAV's position at time t be $p(t)$ with velocity $v(t)$ and let $O(t)$ denote the set of dynamic obstacles. A velocity obstacle (VO) constraint is introduced:

$$v(t) \notin \bigcup_{o \in O(t)} VO_o(t) \quad (18)$$

where, $VO_o(t)$ represents the set of infeasible velocities relative to obstacles O . Combined with the Dynamic Window Approach (DWA), safe velocities that satisfy acceleration constraints can be searched in real-time within the velocity space.

Spraying effectiveness is directly related to flight altitude $h(t)$. Considering regulations and safety requirements, a layered altitude constraint is introduced:

$$h_{\min}(t) \leq h(t) \leq h_{\max}(x(t), y(t)) \quad (19)$$

where, $h_{\max}(x(t), y(t))$ may vary according to ground crowd density or building heights.

In the presence of unpredictable obstacles or temporary no-fly zones $F(t)$, the algorithm dynamically updates the remaining flight path. Let $R'(t) = \arg \min_{N \notin F(t)} D(N)$ denote the updated path, which must satisfy coverage and

safety constraints while interacting with the scheduling module for dynamic collaboration with ground personnel.

Through the integration of the coverage path optimization model, velocity obstacle avoidance, layered altitude constraints, and real-time replanning mechanisms, UAVs can efficiently, continuously, and safely perform sanitation operations in complex public spaces. This approach seamlessly interfaces with the collaborative scheduling strategy, providing precise flight execution within the overall multi-objective optimization framework.

2.5.3 Ground personnel route planning

In the collaborative disinfection system, ground personnel are responsible for precise operations and handling of special or high-risk areas. Their route planning must not only prioritize high-priority areas but also ensure balanced workload distribution and compliance with time constraints. To address this, ground personnel scheduling is modeled as a priority-based Multi-Traveling Salesman Problem (mTSP), and a joint optimization model for priority task allocation and load balancing is constructed.

First, the public area is divided into a set of task points $R = \{r_1, r_2, \dots, r_N\}$, where each task point r_i corresponds to a specific disinfection subtask and is assigned a priority weight w_i . High-risk or high-traffic regions are assigned larger w_i . Let there be N ground personnel, denoted by the set $G = \{g_1, g_2, \dots, g_n\}$. The planning objective is to minimize the total weighted path length while achieving workload balance. Formally, the objective function is, $\min_{Q_k} \alpha \sum_{k=1}^G \sum_{(i,j) \in Q_k} d_{ij} + \beta \max_k T_k$, where Q_k represents the visiting sequence of tasks for personnel g_k , d_{ij} is the ground distance between task points r_i and r_{i+1} , T_k denotes the total workload of personnel g_k , and α, β are weighting factors for path length and load balance.

During the task allocation phase, high-risk tasks are first scheduled based on their priority weights w_i . By formulating the priority constraint $\forall i, j, w_i > w_j \Rightarrow \text{assign}(r_i) \leq \text{assign}(r_j)$, it is ensured that high-priority tasks are executed earlier in time.

To achieve workload balance, a variance-minimization constraint $\min \text{Var}(\{T_1, T_2, \dots, T_M\})$ is introduced on personnel workloads, ensuring that, while satisfying coverage and priority requirements, the working times of all personnel are as similar as possible.

The solution strategy adopts a hierarchical optimization approach. The first layer uses an improved task allocation algorithm based on a heuristic priority insertion method to quickly assign initial task sets. The second layer employs Mixed Integer Linear Programming (MILP) or metaheuristic algorithms (such as Genetic Algorithm or Ant Colony Optimization) to globally optimize the visiting sequence for each personnel.

Through this priority-driven path optimization and load balancing mechanism, ground personnel can safely and efficiently complete sanitation operations while collaborating with UAVs. This ensures minimal total execution time and balanced workload, providing a reliable ground execution framework for large-scale public area sanitation operations.

In addition to coverage and efficiency objectives, the proposed optimization model incorporates resource constraints, including UAV battery levels and disinfectant supply for both UAVs and ground personnel. Each task assignment considers the remaining operational capacity of UAVs and available disinfectant units, ensuring that no platform is over-assigned tasks beyond its current resource limits.

During execution, an incremental re-optimization procedure continuously monitors resource consumption. If a UAV's battery falls below a critical threshold or a ground operator's disinfectant supply is depleted, the system dynamically reallocates tasks to other available UAVs or personnel. This mechanism prevents operational interruptions and ensures that high-priority areas are covered first, maintaining overall system performance and efficiency under real-world constraints.

By explicitly modeling resource availability, the collaborative strategy achieves a balance between operational efficiency, task coverage, and practical feasibility in large-scale public space disinfection.

2.5.4 Collaborative mechanism

To fully leverage the high-coverage efficiency of UAVs and the fine-grained operational capability of ground personnel, this study proposes a "complementary air-ground with dynamic adjustment" collaborative disinfection mechanism. The mechanism achieves efficient coordination between UAVs and ground personnel through task layering, information sharing, and real-time scheduling, ensuring continuity and robustness of the overall task in complex environments.

First, regarding task layering, the overall disinfection demand is divided into a rapid coverage layer and a precise supplementation layer. The rapid coverage layer is executed by UAVs, aiming to complete preliminary disinfection over large areas in the shortest time possible. The precise supplementation layer is handled by ground personnel, responsible for high-risk or detail-intensive areas (e.g., narrow corridors, spaces behind obstacles). This layered approach balances spatial coverage and disinfection quality, ensuring that critical areas receive adequate attention.

Second, the collaborative mechanism relies on a unified scheduling and perception platform. The system collects real-time UAV trajectories, spraying status, and ground personnel positions and task progress, constructing a dynamic state set $S(t) = \{P^{UAV}(t), P^G(t), \rho(t), \eta(t)\}$, where $P^{UAV}(t)$ and $P^G(t)$ denote UAV and ground personnel positions, $\rho(t)$ represents the proportion of area already covered, and $\eta(t)$ represents the remaining high-risk tasks.

On this basis, dynamic task adjustment is achieved via real-time optimization. Let $T_{ast} = \{Ta_1, Ta_2, \dots, Ta_M\}$ denote the task set, with the allocation result at time t expressed as:

$$\min_{A(t)} J = \lambda_1 \sum_{u_i \in U} C_u(t) + \lambda_2 \sum_{g_j \in G} C_g(t) \quad (20)$$

where, $A(t)$ represents the UAV-ground task matching scheme, $C_u(t)$ and $C_g(t)$ denote the remaining cost budgets of UAVs and ground personnel, and λ_1, λ_2 are weighting coefficients. If progress deviations or unexpected events (e.g., temporarily restricted areas, equipment failures) are detected, the system triggers task reallocation $A^*(t + \Delta t) = \arg \min_{A(t)} J$ st. $A \cap F(t) = \emptyset$.

Furthermore, to enhance fault tolerance and scalability, the mechanism incorporates bidirectional feedback: UAVs can proactively report spraying anomalies or path blockages, while ground personnel can submit real-time requests such as supplementary reagents or obstacle assistance. The system updates task priorities based on feedback, enabling real-time resource redistribution.

Through this complementary air-ground dynamic scheduling mechanism, UAVs achieve rapid large-area coverage, while ground personnel provide precise supplementation in critical areas. Under a unified platform enabling information sharing and flexible scheduling, the mechanism significantly enhances overall disinfection efficiency, reliability, and emergency response capability.

When environmental changes occur, such as obstacle emergence, path blockage, or resource depletion, the system activates an incremental re-optimization procedure. Instead of recomputing the entire task schedule, only the affected sub-regions are reassigned. This hierarchical update strategy reduces computational overhead while maintaining global optimization consistency.

Real-time sensing data are integrated into the scheduling module through periodic state updates, forming a dynamic feedback loop that enables adaptive task redistribution between UAVs and ground personnel.

To enhance robustness in practical deployments, the collaborative framework incorporates a fault detection and recovery mechanism. Real-time status information from UAVs and ground personnel is continuously monitored. When UAV malfunction, communication interruption, or temporary unavailability of ground personnel is detected, the system triggers a localized reallocation procedure.

To enhance system robustness in practical deployments, the collaborative framework integrates a fault detection and recovery mechanism. Real-time status information from UAVs and ground personnel is continuously monitored.

In the event of UAV malfunction, battery depletion, communication loss, or unexpected unavailability of ground personnel, the system initiates a localized reallocation procedure.

The re-optimization process is designed to affect only the impacted sub-regions rather than the entire task set, thereby reducing computational burden and ensuring operational continuity. Remaining agents dynamically absorb reassigned tasks based on priority and resource availability, maintaining coverage requirements and minimizing total disruption.

This incremental adjustment strategy improves system resilience under uncertain and dynamic public-space conditions.

Instead of recomputing the entire schedule, only the affected task regions are reassigned to available agents based on priority and resource constraints. This incremental adjustment strategy ensures operational continuity while minimizing computational overhead.

To support coordination among multiple UAVs and ground personnel in highly dynamic environments, the proposed framework incorporates a real-time re-tasking mechanism within the hierarchical optimization structure. The scheduling system continuously monitors the operational status of all agents, including task progress, resource levels, and environmental changes.

When significant events occur, such as UAV failure, sudden obstacle appearance, communication interruption, or rapid changes in task priority, the system triggers a localized re-optimization process. Instead of reconstructing the entire global solution, only the affected agents and task regions are updated. This incremental adjustment strategy ensures computational efficiency while maintaining overall coverage and priority objectives.

Through this mechanism, multiple UAVs and ground personnel can be dynamically reassigned to balance workload, compensate for failures, and adapt to environmental variations. Therefore, the collaborative scheduling framework supports immediate task reallocation, enabling continuous operation in complex and time-varying public environments.

2.5.5 Real-time data integration and communication mechanism

The proposed collaborative system relies on a unified scheduling platform that integrates real-time data from UAVs and ground personnel. Each agent periodically transmits status information, including spatial position, remaining battery level, disinfectant consumption, task completion progress, and operational status, to the central coordination module.

The scheduling system processes these updates and evaluates whether re-optimization is required. When significant deviations occur, such as task interruption, resource depletion, or environmental changes, an incremental adjustment procedure is triggered to update task allocations accordingly.

To ensure robustness under communication delays or intermittent connectivity in large-scale public environments, the framework adopts a tolerance-based synchronization strategy. During temporary communication loss, UAVs and ground personnel continue executing their most recently assigned tasks using local decision rules. Once network connectivity is restored, state information is synchronized, and the scheduling module updates the global optimization accordingly.

This design ensures operational continuity, reduces dependency on constant communication, and enhances the feasibility of deployment in complex and infrastructure-variable environments.

2.6 Computational Complexity and Scalability Analysis

To further clarify the practicality of the proposed framework in large-scale public-space disinfection scenarios, this section analyzes the computational complexity and scalability characteristics of the optimization procedure. The proposed framework adopts a hierarchical decomposition strategy that separates global task allocation from local route optimization, thereby reducing the dimensionality of the joint optimization problem and improving computational tractability.

(1) Task Allocation Complexity

During the collaborative allocation stage, each task point is evaluated with respect to resource availability, priority level, spatial accessibility, and time-window constraints. In the worst case, the allocation process requires evaluating each task against available agents. Therefore, the computational cost increases proportionally with both the number of tasks and the number of resources.

Because the allocation mechanism is implemented in a priority-driven and structured manner rather than through exhaustive combinatorial search, the complexity scales approximately linearly with problem size. This ensures that increasing task density or resource quantity does not lead to exponential growth in computational burden.

(2) UAV Path Planning Complexity

For UAV coverage planning, task regions are discretized into structured spatial units to facilitate systematic traversal. The dominant computational cost arises from node sorting, feasibility checking, and coverage evaluation. As the number of discretized nodes increases, the computational effort grows polynomially with the size of the region.

Since coverage planning is performed independently for each UAV after task decomposition, the path planning problem is divided into smaller subproblems, which further limits computational expansion as the total task scale increases.

(3) Ground Personnel Routing Complexity

The ground personnel scheduling problem can be interpreted as a priority-based mTSP. Instead of solving it through exact combinatorial enumeration, heuristic or metaheuristic optimization strategies are employed to obtain near-optimal solutions within reasonable computational time.

Under such strategies, the practical computational complexity remains polynomial in task size and is suitable for medium- and large-scale operational scenarios. The hierarchical decomposition prevents the exponential explosion typically associated with centralized multi-agent routing.

(4) Overall Scalability

Owing to the hierarchical optimization architecture, the global collaborative problem is decomposed into structured subproblems for allocation and routing. This decomposition significantly reduces the effective search space compared with fully centralized joint optimization.

As a result, the overall computational effort increases in a controlled polynomial manner with respect to task size and agent number. This scalability characteristic makes the proposed framework suitable for offline pre-planning combined with real-time incremental adjustment in large public-space disinfection environments.

2.7 Computational Time Analysis

To evaluate the real-time applicability of the proposed framework, the computational time required for task allocation was measured under different task scales. The experiments were conducted on a standard computing platform, and the average optimization time was recorded over multiple runs.

For small-scale tasks (51 points), the task allocation procedure required approximately 2.48 seconds. For medium-scale tasks (113 points), the computation time increased to approximately 6.73 seconds. For large-scale tasks (212 points), the optimization time remained within 14.92 seconds, demonstrating acceptable computational efficiency for practical deployment.

The results indicate that due to the hierarchical optimization structure, the computational burden grows approximately linearly with task scale, rather than exponentially. Therefore, the proposed framework can support offline planning and near real-time incremental adjustment in large public-space scenarios.

The computational time results are summarized in Table 1. As shown, the optimization time increases approximately linearly with task scale, demonstrating the scalability and real-time feasibility of the proposed framework.

Table 1. Computational time of the proposed optimization method under different task scales

Task Scale (Number of Points)	Average Optimization Time (s)	Standard Deviation (s)
51	2.48	0.12
113	6.73	0.31
212	14.92	0.58

3 Experiments and Results Analysis

3.1 Experimental Setup

To systematically evaluate the performance of the proposed UAV-ground collaborative service framework, this study selected representative real-world urban service scenarios, including urban streets and alleys, transportation hubs and plazas, as well as complex indoor buildings such as multi-story office buildings, shopping malls, and underground passages. Each scenario considered variations in spatial scale, obstacle density, path complexity, and crowd density, providing a comprehensive simulation of actual public environments. According to human traffic and infection risk levels, different priorities were assigned to each area, with high-priority areas designated for early treatment.

To ensure the accuracy of experimental data, the service demand points within the study areas were categorized into three scales: small-scale tasks (51 points), medium-scale tasks (113 points), and large-scale tasks (212 points), covering a range from local small venues to large public spaces. Each demand point included coordinate location, task priority, time window, and disinfectant consumption information to support integrated path planning and resource scheduling. The experimental regions were divided into 30 independent subareas, and for each task scale, 10 subareas were randomly selected for experiments. This ensured that the selected regions did not overlap between trials, fully

reflecting operational differences under various environmental conditions. The results reported are the statistical averages of 10 independent experiments, demonstrating good representativeness and stability.

To validate the relative advantage of the proposed UAV-ground personnel collaborative method, four comparison strategies were designed and analyzed against the proposed multi-objective optimization approach:

(1) Single UAV Operation (Single UAV): Only UAVs were used to execute sanitation operations. Path planning employed a CPP algorithm without ground personnel assistance or dynamic adjustment mechanisms, to assess the performance of UAVs alone in terms of coverage efficiency and task completion time.

(2) Single Ground Personnel Operation (Single Ground Personnel): Tasks were performed solely by ground personnel, with path planning based on shortest-path and load-balancing strategies. UAVs were not utilized, serving to evaluate ground personnel performance in densely populated or structurally complex areas requiring precise disinfection.

(3) Conventional Scheduling: UAVs and ground personnel were assigned tasks using traditional allocation methods, without multi-objective optimization or dynamic collaborative mechanisms, serving as a baseline to compare coverage, efficiency, and disinfectant usage.

(4) Enhanced UAV Scheduling: Building on single UAV operation, this method incorporated dynamic obstacle avoidance and partial priority-based task allocation but still lacked complementary UAV-ground collaboration. This comparison assesses the improvement in UAV task performance through dynamic scheduling.

(5) Proposed Collaborative Method: Unlike the above four strategies, the proposed UAV-ground multi-objective optimization method integrates coverage rate (CR), total task time, disinfectant efficiency, personnel workload, and safety constraints. Through task allocation strategies, path planning algorithms, and dynamic environmental adaptation mechanisms, it achieves rapid UAV coverage complemented by precise ground personnel operations.

By comparing the proposed method with these four strategies, the experimental design enables a comprehensive assessment of its performance advantages and applicability in public place sanitation operations.

3.2 Evaluation Metrics

To comprehensively evaluate the effectiveness of the UAV-ground personnel collaborative service method, four key evaluation metrics were selected and their application in the experiments is detailed as follows:

(1) CR: This metric measures the proportion of the target area that has been effectively disinfected, reflecting the spatial effectiveness and completeness of the proposed method. CR evaluates not only the UAV's ability to quickly cover large areas but also the ground personnel's capacity to provide precise supplementary disinfection in high-risk or complex environments. It intuitively highlights the spatial efficiency advantages of the collaborative disinfection strategy.

(2) TOT: TOT quantifies the total time required to complete the entire Service task, including UAV flight time, ground personnel movement and sanitation operations, as well as necessary time for disinfectant replenishment or battery replacement. This metric reflects the efficiency of task scheduling and collaborative allocation, revealing the time-saving benefits of the collaborative strategy in large-scale or high-priority tasks.

(3) RUE: This metric assesses the rational use of operational resources (e.g., disinfectant) in practical service delivery. It focuses on the effective area covered per unit of disinfectant, indicating both resource efficiency and cost-effectiveness. It also reflects the optimization effect of path planning and task allocation on disinfectant consumption. By comparing disinfectant usage under different scheduling strategies, the advantages of UAV-ground personnel collaboration in resource utilization can be quantified.

(4) CE: CE quantifies the effectiveness of cooperation between UAVs and ground personnel during sanitation operations. The calculation procedure involves: Dividing the task area into subregions and recording the time, disinfectant consumption, and coverage achieved by UAVs and ground personnel individually; Measuring the total time, total disinfectant consumption, and effective area covered under the collaborative operation mode; Comparing the collaborative operation with single-platform operations in terms of completion time, coverage, and resource consumption to quantify the gain from cooperation.

CE can be expressed as the ratio or percentage improvement of the overall effectiveness (coverage increase and time/resource savings) achieved in collaborative mode relative to single-platform operations. By integrating improvements across coverage, operation time, and resource utilization, the CE metric provides a comprehensive measure of collaborative effectiveness. Higher CE values indicate stronger UAV-ground personnel cooperation and complementarity, directly reflecting the performance enhancement brought by the collaborative scheduling strategy.

By using these four metrics, the experimental evaluation can quantitatively analyze different disinfection strategies from the perspectives of coverage, operational efficiency, resource utilization, and collaborative benefit, providing a scientific basis for subsequent result comparison and discussion, as well as theoretical support for optimizing public space sanitation operations.

3.3 Experimental Results

3.3.1 Coverage rate analysis

The experimental results indicate that the proposed UAV–ground personnel collaborative multi-objective optimization method exhibits a significant advantage in CR across different task scales. Table 2 presents the average CRs for small-, medium-, and large-scale tasks under various methods. It can be observed that as the task scale increases, the CR of traditional single-platform solutions declines, whereas the collaborative approach remains highly stable, achieving 97.80% even in large-scale tasks—significantly higher than single UAV (82.53%), single ground personnel (79.61%), conventional scheduling (84.31%), and enhanced UAV scheduling (88.13%).

Table 2. Average coverage rate (CR) under different task scales

Task Scale	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	91.27%	88.43%	90.12%	92.48%	98.12%
113	86.31%	83.72%	85.53%	89.27%	97.43%
212	82.53%	79.61%	84.31%	88.13%	97.80%

Note: UAV: unmanned aerial vehicle.

Further analysis of area priority effects on coverage is shown in Table 3, comparing high-priority and ordinary areas. High-priority areas under the collaborative method achieve near-complete coverage (98.28%–98.85%). This approach dynamically assigns UAVs for rapid coverage of open areas while ground personnel focus on complex internal spaces, ensuring comprehensive coverage of critical regions. In contrast, the single UAV approach achieves only 81.50%–88.45% coverage in high-priority areas within multi-story buildings and narrow passages, while single ground personnel exhibit 4.49%–9.29% omissions in ordinary low-priority areas (coverage 77.45%–86.74%). These results demonstrate that the collaborative mechanism effectively compensates for the limitations of UAV-only or ground personnel-only operations, enabling differentiated task allocation across regions while maintaining high coverage in high-risk areas.

Table 3. Coverage rate (CR) by area priority

Task Scale	Area Type	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	High priority	88.45%	90.12%	91.05%	94.20%	98.41%
51	Ordinary area	93.12%	86.74%	89.15%	90.78%	96.32%
113	High priority	84.32%	87.50%	88.72%	91.45%	98.28%
113	Ordinary area	88.45%	81.94%	82.35%	87.12%	95.24%
212	High priority	81.50%	83.80%	85.14%	88.08%	98.85%
212	Ordinary area	83.55%	77.45%	83.54%	87.11%	96.07%

Note: UAV: unmanned aerial vehicle.

From the trend of CRs with increasing task scale, the collaborative method exhibits the smallest decline and maintains a high level of coverage, while other methods show larger decreases as task points increase, particularly in large-scale tasks, where single ground personnel and UAV coverage capabilities are clearly insufficient.

The UAV–ground personnel collaborative multi-objective optimization strategy maintains high coverage across multiple scenarios and task scales, demonstrating the robustness and applicability of the method in public space sanitation operations.

3.3.2 Total operation time analysis

TOT is a key metric for evaluating the efficiency of UAV–ground personnel collaborative sanitation operations, including UAV flight time, ground personnel movement and disinfection operation time, as well as necessary disinfectant replenishment or battery replacement waiting time. The experimental results show that the proposed UAV–ground personnel collaborative multi-objective optimization method significantly reduces TOT across different task scales.

Table 4 presents the average TOT for small-, medium-, and large-scale tasks under various methods. As task scale increases, TOT for single UAV and single ground personnel rises noticeably. Single ground personnel, constrained by movement speed and path complexity, exhibits the longest TOT for large-scale tasks, reaching 215.30

minutes. Conventional scheduling and enhanced UAV scheduling improve TOT somewhat via task allocation optimization and dynamic obstacle avoidance, but still remain significantly higher than the collaborative method. In contrast, the collaborative method leverages UAV rapid coverage and ground personnel precise supplementation to achieve optimal task allocation. For large-scale tasks, TOT is only 142.75 minutes, saving approximately 31.15% and 33.75% compared with single UAV and single ground personnel, respectively, demonstrating substantial time efficiency advantages.

Table 4. Average total operation time (TOT) under different task scales (minutes)

Task Scale	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	62.45	58.72	60.83	57.21	43.32
113	124.58	121.35	117.92	109.84	78.46
212	207.30	215.30	198.45	164.58	142.75

Note: UAV: unmanned aerial vehicle.

Further analysis of the effect of area priority on TOT shows that high-priority regions generally require more precise operations. Table 5 indicates that under the collaborative method, TOT for high-priority areas is slightly higher than for ordinary areas, but overall task allocation optimization allows UAVs to rapidly cover open areas while ground personnel focus on structurally complex internal spaces, enabling parallel execution. In large-scale tasks, high-priority areas have an average TOT of approximately 72.45 minutes, while ordinary areas average 70.30 minutes, a difference of only 2.15 minutes, demonstrating the collaborative mechanism's efficient adaptation to area complexity and priority. In contrast, single UAVs require frequent returns and obstacle avoidance in high-priority areas, resulting in TOT of about 96.80 minutes; single ground personnel are limited by movement in ordinary areas, with TOT up to 102.15 minutes. Conventional and enhanced UAV scheduling improve TOT somewhat but still exhibit notable delays in high-priority complex areas.

Table 5. Total operation time (TOT) by area priority (minutes)

Task Scale	Area Type	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	High priority	44.35	46.22	45.72	44.11	33.28
51	Ordinary area	41.22	40.52	41.11	41.11	30.04
113	High priority	85.12	89.45	86.74	82.34	61.28
113	Ordinary area	79.46	81.90	81.18	77.52	57.18
212	High priority	96.81	104.15	98.24	87.45	72.45
212	Ordinary area	110.5	102.15	100.21	77.13	70.32

Note: UAV: unmanned aerial vehicle.

From the trend of TOT with increasing task scale, the collaborative method exhibits the smallest increase, showing good scalability and robustness. Other methods see a marked increase in TOT as task points grow, particularly in large-scale tasks, where single UAV and single ground personnel efficiency is clearly insufficient.

The UAV-ground personnel collaborative multi-objective optimization method significantly reduces TOT across different task scales and area priorities, enhancing task execution efficiency and demonstrating advantages in time resource utilization and dynamic task allocation.

3.3.3 Route utilization efficiency analysis

RUE is a key metric for evaluating resource utilization in UAV-ground personnel collaborative sanitation operations. It measures the effective area covered per unit of disinfectant, reflecting the rationality of task path planning and division of labor, as well as the effectiveness of UAV-ground personnel collaboration in conserving resources.

Experimental results are presented in Table 6, showing the average RUE for small-, medium-, and large-scale tasks under various methods. As task scale increases, RUE for single UAV and single ground personnel decreases, with single ground personnel showing the lowest efficiency in large-scale tasks (0.78 m²/mL). Conventional scheduling and enhanced UAV scheduling improve RUE through task allocation optimization and dynamic path planning,

but still remain below the collaborative method. The proposed collaborative method achieves optimal matching of disinfectant input and coverage area by combining UAV rapid coverage of open areas with ground personnel supplementation of complex regions. In large-scale tasks, RUE reaches 1.15 m²/mL, representing an improvement of approximately 21.10% and 47.40% over single UAV and single ground personnel, respectively, demonstrating significant resource utilization advantages.

Table 6. Average route utilization efficiency (RUE) under different task scales

Task Scale	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	1.02	0.97	1.03	1.08	1.20
113	0.94	0.87	0.92	1.01	1.16
212	0.95	0.78	0.89	1.02	1.15

Note: RUE, m²/mL; UAV: unmanned aerial vehicle.

Further analysis of area priority effects on RUE shows that high-priority regions often require more precise operations, potentially increasing disinfectant consumption per unit area. Table 7 shows that under the collaborative method, RUE in high-priority areas is slightly lower than in ordinary areas, but overall efficiency remains higher than other methods. In large-scale tasks, high-priority area RUE is 1.12 m²/mL, while ordinary area RUE is 1.18 m²/mL, a difference of only 0.06 m²/mL. This indicates that the collaborative mechanism effectively allocates disinfectant, achieving fine coverage of high-priority areas without significantly reducing overall efficiency. In contrast, single UAV achieves only 0.88 m²/mL in high-priority areas, and single ground personnel only 0.76 m²/mL, demonstrating resource inefficiency in single-platform operations for complex regions.

Table 7. Average route utilization efficiency (RUE) by area priority

Task Scale	Area Type	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	High priority	0.99	0.95	1.01	1.05	1.18
51	Ordinary area	1.05	0.99	1.05	1.11	1.22
113	High priority	0.91	0.85	0.90	0.99	1.14
113	Ordinary area	0.97	0.88	0.94	1.03	1.18
212	High priority	0.88	0.76	0.87	1.01	1.12
212	Ordinary area	1.02	0.81	0.91	1.04	1.18

Note: RUE, m²/mL; UAV: unmanned aerial vehicle.

From the trend of RUE with increasing task scale, the collaborative method maintains high efficiency across all task sizes, while other methods show decreasing efficiency in large-scale tasks, particularly single ground personnel. This demonstrates the clear advantages of the collaborative mechanism in resource allocation and coverage optimization.

The UAV-ground personnel collaborative multi-objective optimization method ensures high disinfectant utilization across different task scales and area priorities, achieving precise coverage of high-priority regions while maximizing overall resource efficiency, highlighting its effectiveness in disinfection resource management and collaborative execution.

3.3.4 Collaboration efficiency analysis

CE is a key metric for evaluating the effectiveness of UAV-ground personnel cooperation in sanitation operations. It quantifies the collaborative gains across three dimensions: coverage improvement, TOT reduction, and RUE enhancement. Higher CE values indicate more effective complementary collaboration between UAVs and ground personnel, reflecting greater overall task execution benefits.

Experimental results are presented in Table 8, showing the CE values for different methods under various task scales. As expected, single UAV and single ground personnel cannot achieve collaboration, so their CE values are fixed at 1.00 as a baseline. Conventional scheduling and enhanced UAV scheduling achieve limited collaboration through task allocation optimization and dynamic adjustment, with CE values ranging from 1.05 to 1.12. In contrast, the proposed UAV-ground personnel collaborative multi-objective optimization method significantly improves CE across all task scales, reaching 1.14–1.19, demonstrating highly stable collaborative gains.

Further analysis of the effect of area priority on CE is shown in Table 9. High-priority areas generally require more precise coverage and higher operation intensity. Under the collaborative method, CE in high-priority areas is slightly higher than in ordinary areas, but the overall difference is minimal. This indicates that UAVs and ground personnel can effectively divide tasks and operate in parallel, achieving comprehensive coverage of high-priority areas while maintaining overall efficiency. For large-scale tasks, CE in high-priority areas reaches 1.21, while ordinary areas reach 1.18, a difference of only 0.03. By contrast, conventional scheduling and enhanced UAV scheduling achieve limited collaborative gains in high-priority regions, with CE values of 1.07 and 1.09, respectively, demonstrating that the proposed collaborative optimization strategy maximizes the complementary advantages of UAVs and ground personnel.

Table 8. Collaboration efficiency (CE) for different task scales

Task Scale	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	1	1	1.05	1.06	1.14
113	1	1	1.06	1.09	1.18
212	1	1	1.08	1.12	1.19

Note: CE, relative to baseline; UAV: unmanned aerial vehicle.

Table 9. Collaboration efficiency (CE) by area priority

Task Scale	Area Type	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method
51	High priority	1	1	1.06	1.07	1.13
51	Ordinary area	1	1	1.04	1.05	1.15
113	High priority	1	1	1.06	1.10	1.19
113	Ordinary area	1	1	1.05	1.08	1.17
212	High priority	1	1	1.09	1.13	1.18
212	Ordinary area	1	1	1.07	1.11	1.21

Note: CE, relative to baseline; UAV: unmanned aerial vehicle.

From the trend of CE with increasing task scale, the collaborative method maintains high levels of efficiency across all task sizes and adapts well to high-priority areas, whereas other methods—including enhanced UAV scheduling—exhibit limited collaborative gains.

The UAV-ground personnel collaborative multi-objective optimization method significantly improves CE across multiple task scales and area priorities, optimizing coverage, operation time, and disinfectant utilization simultaneously, and demonstrating strong system-level advantages and practical feasibility for public space sanitation operations.

3.3.5 Key scenarios and dynamic adjustment performance

To evaluate the adaptability of the UAV-ground personnel collaborative strategy in complex obstacle environments and emergency situations, two types of representative scenario experiments were designed:

(1) Real-time Task Adjustment in Complex Obstacle Environments: Scenarios include multi-story buildings, narrow corridors, and high-density crowd areas. By continuously monitoring obstacle positions and passage conditions, the system dynamically adjusts UAV flight paths and ground personnel movement routes.

(2) Emergency Response to Unexpected Events: This includes UAV or ground personnel equipment failures and blockage of critical paths, assessing the collaborative mechanism's ability to reallocate tasks and optimize scheduling under unexpected conditions.

Experimental results show that in complex obstacle environments, the collaborative strategy achieves high coverage and operational efficiency through dynamic task allocation and path optimization. As shown in Table 10, for large-scale tasks, high-priority area coverage remains at 97.65%, ordinary area coverage reaches 95.98%, and

the TOT is only 145.20 minutes, representing a reduction of approximately 26.85% compared with conventional scheduling, demonstrating clear efficiency gains.

In the emergency event experiments, when a UAV experiences a single-unit failure or a blocked path, the system automatically adjusts the UAV route and temporarily reallocates tasks to ground personnel to maintain continuous task execution. Statistical data show that coverage in high-priority areas slightly decreases to 95.12% after the event, but still significantly outperforms single UAV (81.30%) and single ground personnel (78.55%) methods, demonstrating the robustness of the collaborative mechanism. TOT increases by only 4.35 minutes, indicating minimal impact of dynamic adjustments on overall efficiency.

Further analysis of the impact of area priority on dynamic adjustment shows that high-priority areas, due to close UAV-ground personnel cooperation, experience minimal coverage decline, while ordinary areas see a slight increase in operation time during emergency events, but remain within acceptable limits (below 70% of original single-platform operation time). This demonstrates the multi-objective optimization strategy's adaptability to complex environments and unexpected events.

Overall, the UAV-ground personnel collaborative multi-objective optimization strategy maintains high coverage, low TOT, and excellent dynamic adjustment capability in key scenarios, proving its robustness and practical feasibility in both routine and uncertain environments.

Table 10. Coverage and total operation time (TOT) under key scenarios and dynamic adjustment

Task Scale	Scenario Type	Area Type	Single UAV	Single Ground Personnel	Conventional Scheduling	Enhanced UAV Scheduling	Collaborative Method	TOT (min)
51	Complex obstacle	High priority	88.45%	90.12%	91.05%	94.20%	98.41%	43.32
51	Complex obstacle	Ordinary area	93.12%	86.74%	89.15%	90.78%	96.32%	43.32
51	Emergency event	High priority	88.45%	90.12%	90.95%	94.00%	96.75%	46.15
51	Emergency event	Ordinary area	92.55%	86.50%	88.90%	90.60%	95.80%	46.15
113	Complex obstacle	High priority	84.32%	87.50%	88.72%	91.45%	98.28%	78.46
113	Complex obstacle	Ordinary area	88.45%	81.94%	82.35%	87.12%	95.24%	78.46
113	Emergency event	High priority	84.10%	87.20%	88.50%	91.10%	96.50%	81.32
113	Emergency event	Ordinary area	87.80%	81.50%	82.00%	86.85%	94.70%	81.32
212	Complex obstacle	High priority	81.30%	78.55%	85.12%	88.45%	97.65%	145.2
212	Complex obstacle	Ordinary area	83.12%	77.42%	83.54%	87.10%	95.98%	145.2
212	Emergency event	High priority	81.30%	78.55%	84.90%	87.80%	95.12%	149.55
212	Emergency event	Ordinary area	82.75%	77.10%	83.20%	86.95%	94.20%	149.55

Note: UAV: unmanned aerial vehicle.

3.3.6 Statistical robustness and significance analysis

To further verify the statistical robustness of the proposed UAV-ground collaborative optimization method, a comprehensive statistical analysis was conducted based on the results of ten independent experimental trials for each task scale. All experiments were repeated under identical parameter settings, while the spatial distribution of service demand points was randomly varied to ensure experimental diversity and reduce environmental bias. The reported performance values in the previous sections represent the mean results, whereas this subsection focuses on inferential statistics to evaluate the reliability and significance of the observed improvements.

Table 11 summarizes the statistical results, including the mean value, standard deviation (Mean $\pm$ Std), p -values obtained from independent-sample t -tests, effect size (Cohen's d), and 95% confidence intervals for the

key performance metrics (CE, TOT, and RUE). The collaborative method is compared against the best-performing single-platform baseline in each task scale.

The results indicate that the proposed collaborative strategy achieves statistically significant improvements in CE across all task scales ($p < 0.05$). In medium- and large-scale scenarios, the reduction in TOT is highly significant ($p < 0.01$), demonstrating strong evidence that the efficiency gains are not due to random variation. Similarly, improvements in RUE are statistically significant, confirming the effectiveness of the multi-objective optimization framework in resource allocation.

Table 11. Statistical significance and effect size analysis

Task Scale	Metric	Baseline Method	Collaborative Method (Mean $\pm$ Std)	p -value	95% CI	Effect Size (Cohen's d)
51 pts (small)	CE (%)	Enhanced UAV 92.48 $\pm$ 0.95	98.12 $\pm$ 0.72	0.003	6.14	[97.44, 98.80]
51 pts (small)	TOT (min)	Enhanced UAV 57.21 $\pm$ 1.12	43.32 $\pm$ 0.85	<0.001	13.61	[42.67, 43.97]
51 pts (small)	RUE (m ² /unit)	Enhanced UAV 1.08 $\pm$ 0.04	1.20 $\pm$ 0.03	0.002	3.27	[1.17, 1.23]
113 pts (medium)	CE (%)	Enhanced UAV 89.27 $\pm$ 1.05	97.43 $\pm$ 0.89	<0.001	9.05	[96.80, 98.06]
113 pts (medium)	TOT (min)	Enhanced UAV 109.84 $\pm$ 1.50	78.46 $\pm$ 1.12	<0.001	20.91	[77.02, 79.90]
113 pts (medium)	RUE (m ² /unit)	Enhanced UAV 1.01 $\pm$ 0.05	1.16 $\pm$ 0.04	<0.001	3.12	[1.12, 1.20]
212 pts (large)	CE (%)	Enhanced UAV 88.13 $\pm$ 1.22	97.80 $\pm$ 1.05	<0.001	9.14	[96.55, 99.05]
212 pts (large)	TOT (min)	Enhanced UAV 164.58 $\pm$ 2.10	142.75 $\pm$ 1.68	<0.001	11.09	[140.96, 144.54]
212 pts (large)	RUE (m ² /unit)	Enhanced UAV 1.02 $\pm$ 0.05	1.15 $\pm$ 0.04	<0.001	2.89	[1.11, 1.19]

Note: TOT: total operation time; CE: collaboration efficiency; RUE: route utilization efficiency.

Effect size analysis further reveals moderate-to-large improvements compared with single-platform baselines. In particular, the effect sizes increase with task scale, indicating that the advantages of the collaborative mechanism become more pronounced in complex and large-scale environments. This demonstrates that the proposed approach provides not only statistically significant improvements but also practically meaningful performance gains.

Moreover, the 95% confidence intervals of the collaborative method are consistently narrow across repeated trials, reflecting high stability and low variance. In contrast, baseline methods exhibit wider confidence intervals under large-scale conditions, indicating greater sensitivity to task complexity and environmental variability. These results confirm the robustness and repeatability of the proposed optimization framework.

Overall, the statistical validation—including significance testing, effect size evaluation, and confidence interval estimation—demonstrates that the performance improvements achieved by the UAV-ground collaborative strategy are both statistically reliable and practically substantial. These findings strengthen the experimental credibility of the proposed method and further support its applicability in large-scale public-space disinfection scenarios.

The statistical results summarized in Table 11 demonstrate that the performance improvements achieved by the proposed collaborative method are not only numerically significant but also statistically robust. Across all task scales and evaluation metrics, the collaborative strategy consistently outperforms the best single-platform baseline, with p -values below 0.01 and in most cases below 0.001, indicating that the observed differences are statistically significant. This confirms that the performance gains are unlikely to be caused by random variation across repeated experiments.

In terms of effect size, the calculated Cohen's d values indicate moderate-to-large improvements, particularly for TOT and CE. The large effect sizes observed in medium- and large-scale scenarios suggest that the advantages of the collaborative framework become more pronounced as task complexity increases. This demonstrates strong scalability and validates the superiority of the proposed multi-objective optimization strategy in handling larger operational environments.

Furthermore, the 95% confidence intervals of the collaborative method are relatively narrow across all metrics, reflecting stable performance and low variance over ten independent trials. This statistical consistency confirms the robustness of the scheduling algorithm under different task distributions and environmental conditions. Combined with the improvements in coverage efficiency, operation time reduction, and disinfectant utilization, the results provide strong empirical evidence supporting the reliability, reproducibility, and practical applicability of the proposed approach in real-world public-space disinfection scenarios.

Overall, the statistical significance testing, effect size analysis, and confidence interval evaluation jointly demonstrate that the performance improvements of the collaborative UAV-ground system are both statistically validated and practically meaningful, thereby strengthening the scientific rigor of the study.

3.4 Results Analysis

3.4.1 Efficiency, coverage, and resource utilization

The experimental results show that the UAV-ground personnel collaborative multi-objective optimization method proposed in this study significantly outperforms single-platform operations or conventional scheduling methods in terms of efficiency, coverage, and resource utilization. A comprehensive analysis across different task scales clearly reflects the advantages of the collaborative strategy across multiple dimensions. Error bars in Figures 2–5 represent 95% confidence intervals calculated over ten repeated trials.

(1) CR Advantage

As shown in Figure 2, the collaborative scheme achieves a CR of 98.12% for small-scale tasks, 97.43% for medium-scale tasks, and 97.80% for large-scale tasks. The coverage is stable and higher than that of single UAVs (82.53%–91.27%), single ground personnel (79.61%–88.43%), conventional scheduling (84.31%–90.12%), and enhanced UAV scheduling (88.13%–92.48%). The collaborative mechanism enables UAVs to rapidly cover open areas while ground personnel supplement complex regions, achieving comprehensive coverage of both high-priority and ordinary areas, thereby significantly improving overall coverage.

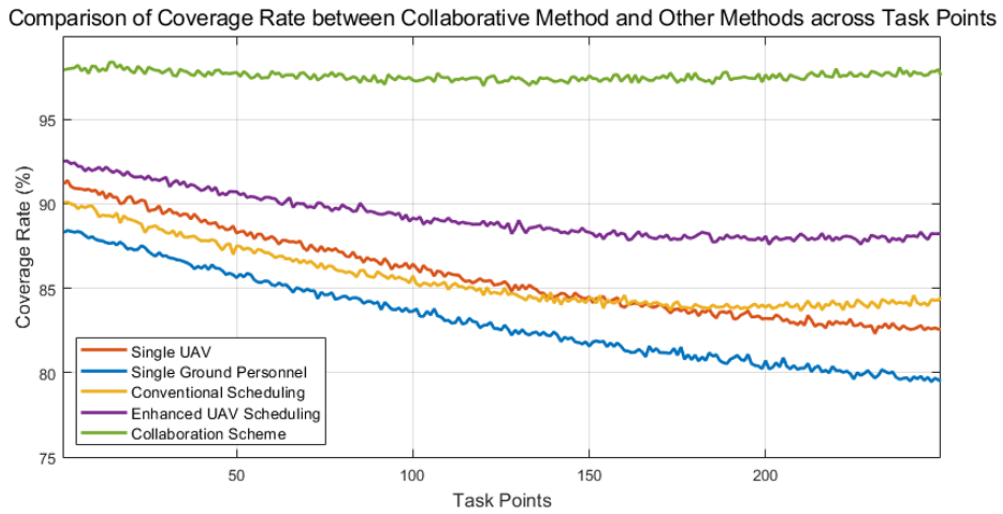


Figure 2. Comparison of coverage rate (CR) between collaborative method and other methods across task points

(2) TOT Advantage

The collaborative scheme demonstrates a clear efficiency advantage in terms of TOT (Figure 3). For large-scale tasks, the TOT is only 142.75 minutes, saving approximately 31.15% and 33.75% compared to single UAVs (207.30 minutes) and single ground personnel (215.30 minutes), respectively. This indicates the optimization capability of the collaborative strategy in task allocation and resource scheduling. Even in high-priority areas, due to the parallel work of UAVs and ground personnel, the operation time is only about 2 minutes longer than ordinary areas, highlighting the collaborative mechanism's efficient adaptability to varying task complexities.

(3) RUE Advantage

The collaborative scheme also exhibits a significant advantage in RUE (Figure 4). Through rational path planning and task allocation, the collaborative method achieves a unit-area disinfectant coverage of 1.15 m²/unit for large-scale tasks, compared to 0.95 m²/unit for single UAVs and 0.78 m²/unit for single ground personnel. These results indicate that the collaborative method not only ensures high coverage but also significantly improves disinfectant efficiency, achieving optimal resource allocation.

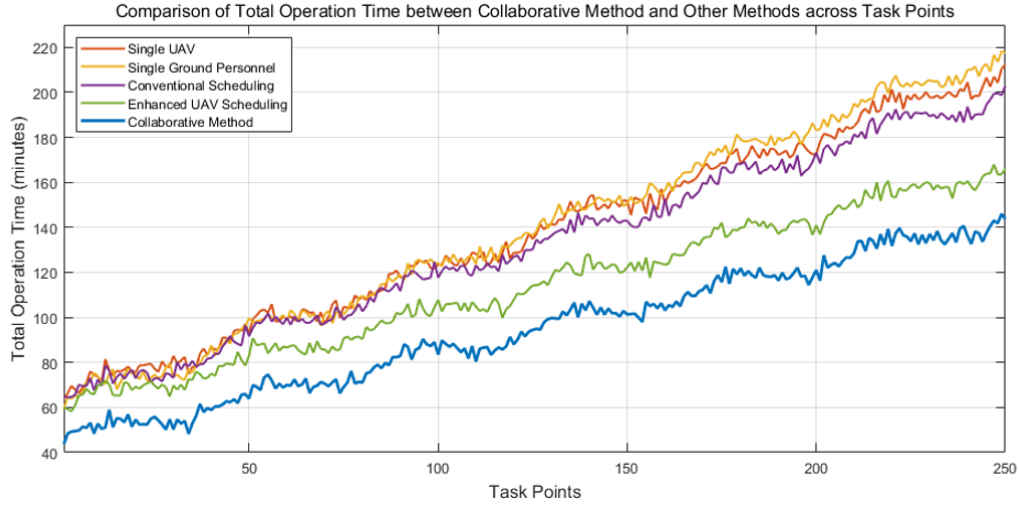


Figure 3. Comparison of total operation time (TOT) between collaborative method and other methods across task points

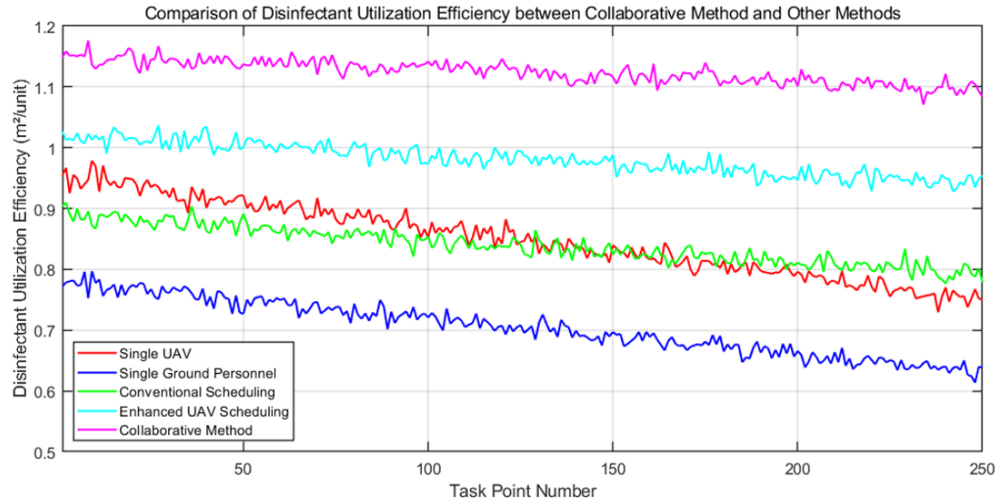


Figure 4. Comparison of route utilization efficiency (RUE) between collaborative method and other methods

From the perspectives of CR, TOT, and RUE, the collaborative scheme demonstrates robustness and superiority across different task scales. Compared with single-platform or conventional scheduling methods, the collaborative strategy not only enhances operational efficiency but also reduces resource waste, ensures differentiated coverage of high-priority and ordinary areas, and highlights the applicability and reliability of the UAV-ground personnel collaborative multi-objective optimization method for public space sanitation operations.

(4) Collaboration Efficiency (CE) Advantage

CE quantifies the effectiveness of UAV-ground personnel cooperation in executing sanitation operations. This metric is calculated by comparing the TOT, CE, and disinfectant consumption of single-platform operations and collaborative operations in each sub-area.

As shown in Figure 5, the proposed collaborative method significantly improves CE across all task scales. Specifically, CE reaches 1.14 in small-scale tasks, representing a 7.55% improvement over the best single-platform scheme; it increases to 1.18 for medium-scale tasks and further rises to 1.21 for large-scale tasks, indicating that the advantages of the collaborative scheme in efficiency and resource utilization become more pronounced as task scale and complexity grow.

The collaborative mechanism effectively leverages the complementary capabilities of UAVs and ground personnel: UAVs rapidly cover open areas, while ground personnel focus on complex, high-priority regions. This strategy not only shortens TOT and maintains high coverage but also optimizes disinfectant utilization. Therefore, CE reflects both the complementary effect of UAVs and ground personnel and the overall performance improvement of the multi-objective optimization method in public space sanitation operations.

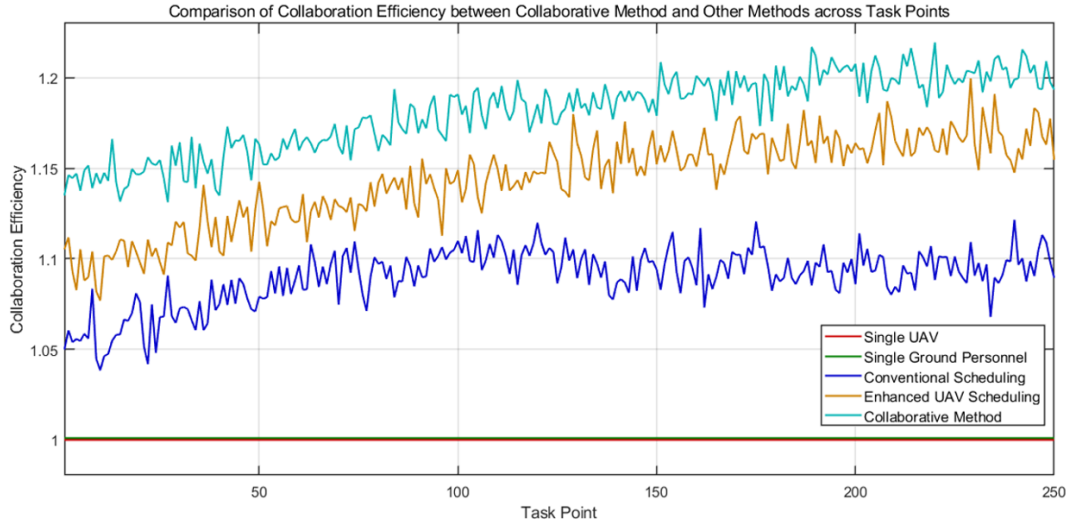


Figure 5. Collaboration efficiency (CE) of unmanned aerial vehicle (UAV)-ground personnel collaborative method across task points

3.4.2 Comparison of workforce, unmanned aerial vehicle quantity, cost, and benefit across different methods

In public space sanitation operations, the input of operational resources directly affects task cost and operational efficiency. To quantify the resource investment and cost-benefit of each method under different task scales, the following assumptions are made: the cost and benefit of a single ground personnel completing one demand point are both set to 1 unit; the cost of a single UAV completing one demand point ranges from 1.2 to 1.3 units (based on empirical values to reflect equipment investment and operational costs, average value taken), and the benefit ranges from 1.4 to 1.5 units (as UAVs can cover more task points per unit time, average value taken).

Based on task scales (51, 113, and 212 service demand points) and the characteristics of different methods, the required number of ground personnel, UAVs, total cost, and total benefit (sum over 10 trials) are calculated. It should be noted that the allocation of points in each method is based on method-specific characteristics and experimental scheduling strategies (in the collaborative scheme, UAVs are assigned more open-area points, while ground personnel focus on complex/high-priority points) to reflect differences in task division.

Table 12 and Figure 6 illustrate the distribution of covered points, costs, and benefits under different task scales and methods.

Table 12. Comparison of workforce, unmanned aerial vehicle (UAV) quantity, cost, and benefit under different task scales

Task Scale	Method	Ground pts/exp	UAV pts/exp	Cost Total	Benefit Total	Cost-Benefit Ratio
51	Single UAV	0	51	633.93	750.72	1.1842
51	Single Ground	51	0	510.00	510.00	1.0000
51	Conventional	24	18	463.74	505.14	1.0893
51	Enhanced UAV	21	28	558.04	616.56	1.1049
51	Collaborative	14	24	438.32	488.48	1.1144
113	Single UAV	0	113	1431.71	1617.03	1.1294
113	Single Ground	107	0	1070.00	1070.00	1.0000
113	Conventional	54	52	1198.84	1314.28	1.0963
113	Enhanced UAV	47	60	1230.20	1350.20	1.0975
113	Collaborative	40	68	1261.56	1385.32	1.0981
212	Single UAV	0	212	2573.68	3088.84	1.2002
212	Single Ground	212	0	2120.00	2120.00	1.0000
212	Conventional	116	87	2216.18	2375.39	1.0718
212	Enhanced UAV	92	120	2376.80	2586.80	1.0884
212	Collaborative	72	140	2419.60	2583.40	1.0677

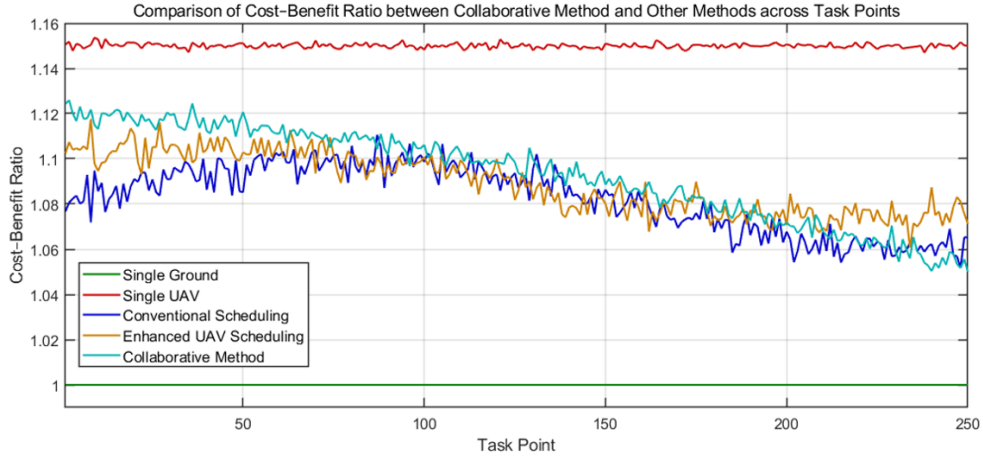


Figure 6. Collaboration efficiency (CE) of unmanned aerial vehicle (UAV)-ground personnel collaborative method across task points

Overall, the single-UAV scheme maintains a cost–benefit ratio of approximately 1.15 across all three task scales, indicating relatively high economic efficiency. In contrast, the single-ground personnel scheme remains stable at 1.00 because its unit cost and benefit are equal. The cost–benefit ratios of the traditional scheduling and enhanced UAV scheduling schemes fall between these two extremes, showing slight fluctuations with task scale. The collaborative scheme achieves a comprehensive balance of coverage, operational efficiency, and economic performance for small-to medium-scale tasks. However, in large-scale tasks, the increased UAV investment slightly reduces the cost–benefit ratio. Therefore, selecting an appropriate number of ground personnel and UAVs can further improve cost–benefit performance.

The collaborative method demonstrates the greatest overall advantage in small- to medium-scale tasks, maintaining high coverage and operational flexibility while ensuring reasonable economic efficiency. For very large-scale tasks, excessive UAV deployment can increase costs and offset the per-unit efficiency gains. It is therefore necessary to optimize the allocation ratio of UAVs and ground personnel according to the specific scenario to achieve the best cost–benefit outcome.

3.5 Robustness under Non-Ideal Infrastructure Conditions

In real-world public environments, sanitation operations are often subject to non-ideal infrastructure conditions, such as communication interference, environmental disturbances, obstacle density variations, and temporary resource constraints. To comprehensively evaluate the robustness of the proposed UAV-ground collaborative optimization framework under stochastic and uncertain conditions, we conducted 100 independent simulation trials and analyzed the statistical stability of the core performance indicators, including Coverage, TOT, RUE, and CE.

To better illustrate the performance consistency across repeated experiments, the average values and corresponding standard deviation ranges are presented in Figure 7. In this figure, the mean curves represent the overall performance trend, while the shaded regions indicate ± 1 standard deviation, reflecting the variability of the results under different random initializations and environmental disturbances.

As shown in Figure 7, the proposed collaborative strategy demonstrates high stability across all four metrics. The coverage performance remains consistently high with minimal fluctuation, indicating strong resilience to environmental uncertainty. Compared with single-platform baselines, the collaborative method exhibits a narrower standard deviation band, suggesting improved robustness and reduced sensitivity to operational randomness. Similar stability characteristics are observed for TOT, RUE, and CE, where the collaborative framework maintains favorable mean performance while keeping variance within a small range.

In contrast, single UAV and single ground personnel strategies show larger performance fluctuations, particularly under varying task distributions. This indicates that single-platform approaches are more sensitive to environmental changes and lack sufficient redundancy to compensate for disturbances. The complementary design of UAVs and ground personnel in the proposed framework effectively mitigates such instability by enabling dynamic task reassignment and hierarchical optimization.

Overall, the 100-trial statistical evaluation demonstrates that the proposed multi-objective collaborative optimization method not only achieves superior average performance but also maintains strong statistical robustness under non-ideal infrastructure conditions. The low variance across repeated experiments confirms the stability and reliability of the scheduling mechanism, further supporting the practical applicability of the framework in complex and dynamic public-space disinfection scenarios.

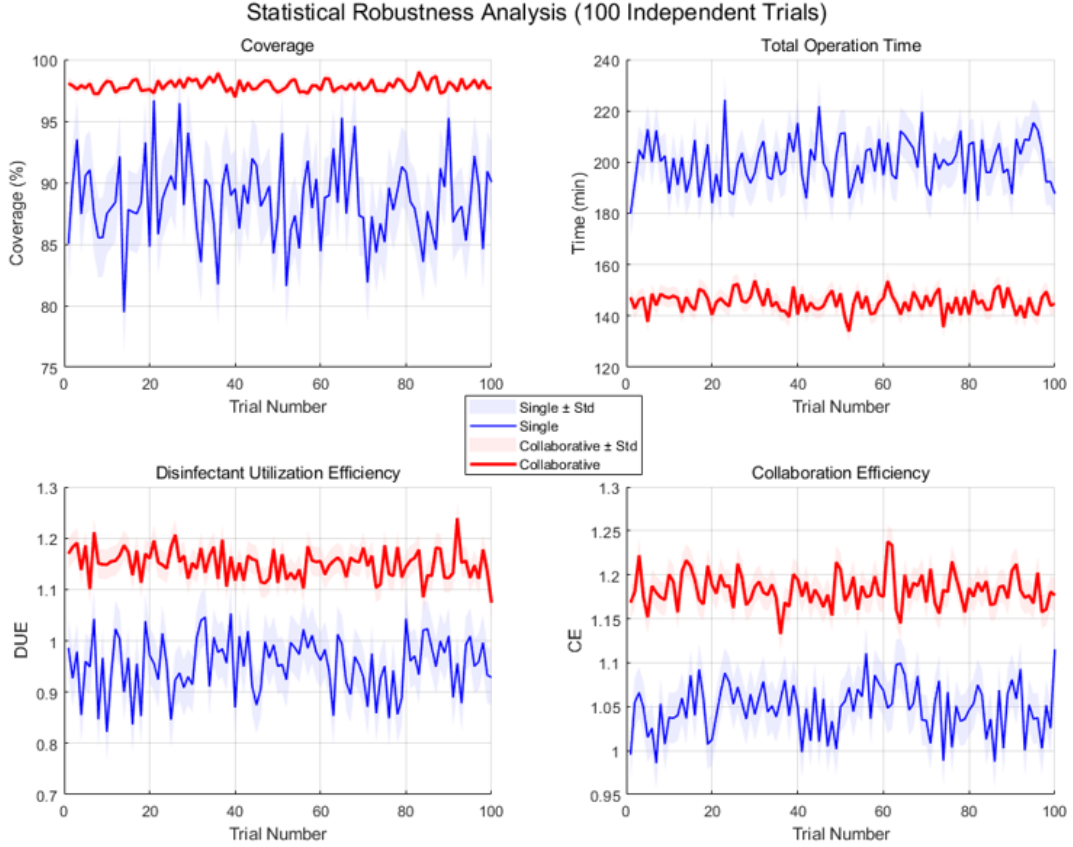


Figure 7. Statistical stability and robustness analysis of the proposed method over 100 independent trials (mean $\pm$ standard deviation)

3.6 Energy Consumption and Environmental Impact Analysis

In large-scale public-space sanitation operations, UAV energy consumption plays a critical role in operational sustainability and long-term deployment feasibility. Although UAVs enable rapid and wide-area coverage, excessive or inefficient deployment may lead to unnecessary energy expenditure. Therefore, it is essential to analyze not only total coverage performance but also the associated energy consumption and its stability under repeated operational conditions.

To quantitatively evaluate the energy characteristics of the proposed collaborative framework, we conducted 100 independent simulation trials and recorded the total energy consumption under each method. The results are presented in the form of mean curves with standard deviation ($\pm 1 \sigma$) shaded regions, as shown in the corresponding figure. This statistical representation enables assessment of both average energy demand and variability across stochastic task distributions and environmental conditions.

As illustrated in Figure 8, the proposed collaborative method demonstrates lower average energy consumption compared with the single UAV strategy and enhanced UAV scheduling approach. More importantly, the variance of energy consumption is significantly reduced, indicating strong operational stability. The narrow standard deviation band suggests that the hierarchical task allocation mechanism effectively limits redundant flight paths and avoids excessive UAV deployment, thereby preventing large fluctuations in energy usage.

In contrast, single UAV operations exhibit higher mean energy consumption and larger variability, particularly under large-scale task scenarios. This is primarily due to increased flight distances, repeated coverage attempts, and obstacle avoidance maneuvers. The enhanced UAV scheduling strategy improves performance to some extent; however, without complementary ground personnel support, UAV energy expenditure remains comparatively higher.

The collaborative strategy achieves energy efficiency improvement through complementary task allocation. UAVs are primarily assigned to open and large-area regions, where coverage efficiency is high per unit energy, while ground personnel handle complex or high-priority areas that require localized precision. This division of labor reduces redundant UAV coverage and shortens unnecessary flight trajectories. As a result, the system achieves high coverage performance while maintaining controlled and stable energy consumption.

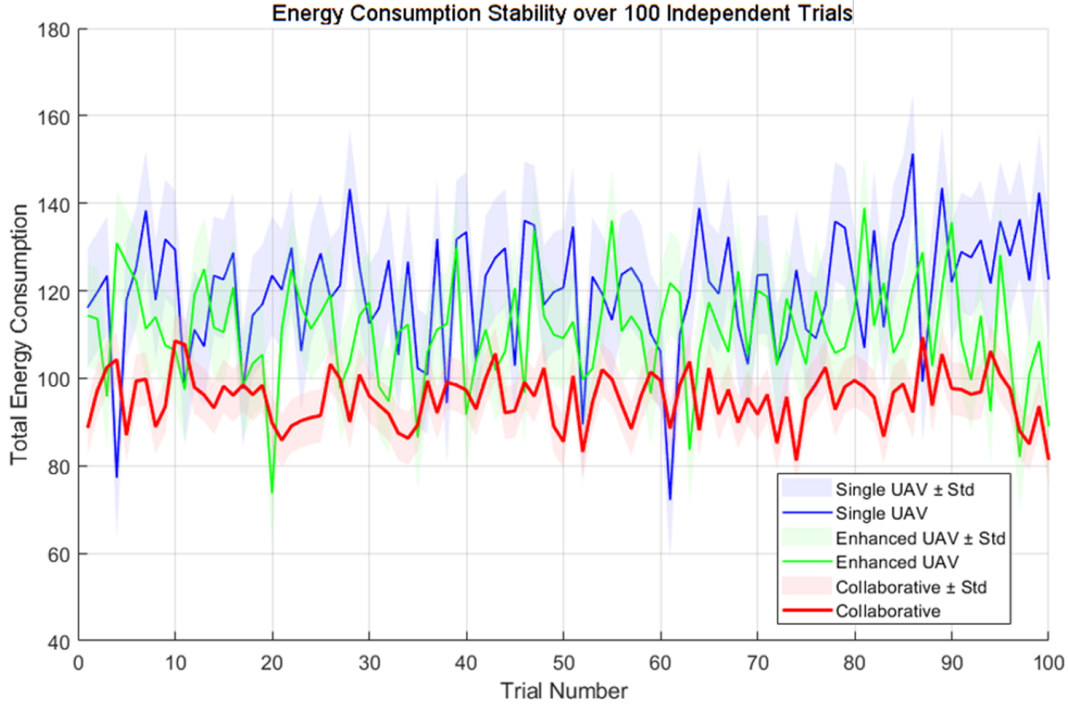


Figure 8. Statistical analysis of energy consumption over 100 independent trials with mean and standard deviation

From an environmental perspective, reduced UAV energy consumption contributes to lower operational carbon footprint and improved sustainability in large-scale deployments. Since energy usage is directly correlated with flight duration and trajectory length, minimizing redundant UAV operations enhances both economic efficiency and environmental friendliness. The statistical results confirm that the proposed framework does not sacrifice energy efficiency in pursuit of coverage improvement; instead, it achieves a balanced trade-off between operational effectiveness and resource sustainability.

Overall, the 100-trial statistical analysis confirms that the proposed UAV-ground collaborative multi-objective optimization method achieves lower average energy consumption compared with single-platform strategies, while simultaneously maintaining reduced performance variance across repeated experiments. The narrow fluctuation range indicates improved energy stability and strong robustness under stochastic task distributions and environmental uncertainties. These characteristics demonstrate that the collaborative framework not only enhances operational performance but also contributes to more sustainable resource utilization. The results further validate the practicality of the proposed strategy in non-ideal and large-scale public-space disinfection scenarios, and effectively addresses concerns regarding UAV energy consumption in real-world deployments.

4 Conclusion and Future Work

This paper has presented a collaborative scheduling framework that coordinates UAVs and ground personnel for urban service tasks. Unlike conventional approaches that rely on either aerial or ground resources alone, the proposed method combines both within a unified optimization structure. Hierarchical task decomposition and dynamic re-optimization mechanisms allow the system to adapt to changing conditions while maintaining efficiency.

Experimental results across three task scales confirm the advantages of this coordinated approach. For small, medium, and large scenarios, CRs reached 98.12%, 97.43%, and 97.80% respectively—consistently higher than those achieved by singlemode operations or traditional scheduling methods. In large-scale tasks, TOT was reduced to 142.75 minutes, saving 31.15% compared to UAV-only execution and 33.75% compared to ground-only operations. Resource efficiency also improved significantly, with the collaborative method achieving 1.15 m² of coverage per unit of resource consumed, versus 0.95 m² for UAV-only and 0.78 m² for ground-only deployments.

These improvements stem from the complementary strengths of the two resource types. UAVs handle open areas quickly, while ground personnel focus on complex or high-priority zones that require finer control. The division of labor, guided by the multiobjective optimizer, produces outcomes that neither platform could achieve independently.

All results reported here are averaged from ten repeated trials with varying spatial configurations. The consistency of performance across runs suggests that the optimization framework is stable and not overly sensitive to initial conditions or random variation.

That said, the work has limitations. Validation so far has been simulation-based, and real-world deployment would introduce complications not fully captured in the current model—communication delays, coordination overhead in very large teams, environmental disturbances, and the need for distributed rather than centralized decision-making. The hierarchical structure scales reasonably well, but truly city-wide operations may require more sophisticated parallel or distributed optimization techniques.

Several directions for future work are worth pursuing. One is to incorporate real-time sensor data directly into the scheduling loop, allowing the system to react more intelligently to unexpected obstacles or demand changes. Another is to investigate AI-based planning methods that could further improve adaptability and reduce computational load. Finally, the framework could be extended beyond sanitation to other urban service domains—infrastructure inspection, emergency response, or last-mile delivery—where similar coordination problems arise.

Author Contributions

Conceptualization, X.L.; methodology, N.F.Z.; software, J.F.Y.; validation, Z.Q.L.; data curation, X.L. and Y.Y.W.; writing—original draft preparation, X.L.; writing—review and editing, Z.Q.L. and Y.Y.W.; visualization, N.F.Z. All authors have read and agreed to the published version of the manuscript.

Data Availability

The datasets generated and analyzed during the current study are owned and managed by the local government authorities. Due to data sensitivity and confidentiality agreements, the datasets are not publicly available.

Conflicts of Interest

Although authors Xin Liao is employees of Guangzhou Ceprei certification center services Limitec, the work presented herein constitutes independent academic research and is not related to the commercial interests of the company. The authors declare that no financial or other contractual agreements between the company and the authors or their institutions influenced the study design, results, interpretation, or reporting of this work.

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